Hybrid Semantic Compiler Architectures

Frontier Semantic Operators, Governed Closure, and the Moving Boundary Between Fuzzy and Deterministic Semantic Computation

LLMs Are Semantic Scouts, Not AGI Oracles

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Abstract

The term "large language model" describes an important technical lineage, but it increasingly misclassifies the operational role of contemporary AI systems. Systems called LLMs now perform research assistance, coding, tool selection, retrieval orchestration, summarization, design exploration, semantic classification, explanation, planning, and interface mediation. These behaviors are better understood as forms of semantic computation under uncertainty.

This paper proposes a refined taxonomy: Fuzzy Semantic Compiler (FSC), Frontier Semantic Operator (FSO), Deterministic Semantic Compiler (DSC), and Hybrid Semantic Compiler (HSC). An FSC is a probabilistic semantic engine operating over underspecified, partially bound, or high-entropy material. In its strongest architectural role, the FSC functions as an FSO: a semantic scout operating where governed closure has not yet been achieved. A DSC is not merely a compiler over typed intermediate representations. It is a deterministic semantic substrate over governed state: a family of bounded, reproducible, governable, and receiptable operators that stabilize, bind, retrieve, route, validate, traverse, execute, record, admit, refuse, or re-present semantic material. An HSC is the governed loop in which FSC/FSO activity and DSC substrate operation cooperate across a moving boundary of closure.

The central thesis is that HSC is not a static pipeline in which FSC proposes and DSC disposes. It is a moving-frontier architecture. FSCs scout regions where semantic architecture is incomplete. Repeated useful FSC maneuvers may expose latent semantic structures, but recurrence alone is not proof of architecture; it may also reveal bias, habit, prompt convention, or anti-pattern. DSCs stabilize only those structures that survive binding, testing, authority review, continuity checks, refusal conditions, and receipts. The frontier advances through crystallization when fuzzy semantic skill becomes deterministic substrate, and retreats through decrystallization when substrate fails rot tests, loses authority, becomes stale, or

no longer satisfies governed closure.

The goal is not to replace FSCs with DSCs, nor to preserve FSCs as permanent black-box authorities. The goal is FSC in its right place: a frontier semantic operator surrounded by a hyper-capable deterministic semantic environment that can also be operated by humans.

1. Introduction

The term "large language model" names a training lineage and interface surface, not the full architecture now emerging. It describes systems trained over large corpora to predict and transform token sequences, but it does not fully describe what these systems now do inside practical AI applications. Contemporary frontier AI systems search, retrieve, code, summarize, plan, reason, classify, invoke tools, inspect documents, compose evidence, generate schemas, and help humans design new conceptual machinery.

The resulting system is not simply a language model. It is a semantic engine operating inside a larger computational environment.

This paper proposes that modern AI should be understood through the architecture of the Hybrid Semantic Compiler. The word "compiler" is used broadly and deliberately. In ordinary programming-language contexts, a compiler is expected to preserve specified semantics while translating from one representation to another. FSCs do not guarantee that kind of preservation. They are not compilers in the verified-translation sense. The term is used here in the weaker architectural sense of representation-transforming machinery operating under constraints. Where this usage risks confusion, FSO names the stronger and more precise role: frontier semantic operation before governed closure exists.

The most important correction is that the FSC should not be reduced to a mere "proposal component." FSC outputs are proposals at jurisdiction boundaries: when they seek entry into memory, evidence, authority, execution, publication, or durable state. But the FSC itself is more than a proposal emitter. In its strongest role, it is a Frontier Semantic Operator: a semantic scout capable of operating productively where no admitted deterministic semantic architecture yet exists.

The core frame is:

```
FSC scouts.
DSC stabilizes.
HSC crystallizes.
```

This slogan is not a replacement for formal architecture. It names the movement of the system. The FSC operates at the warm edge of meaning, where terms may be unstable, relevant models may be missing, authority roots may be unclear, and salience may not yet be declared. The DSC operates where sufficient structure exists to bind, traverse, check, admit, refuse, record, and replay. The HSC is

the governed architecture that moves semantic work from frontier operation into deterministic substrate where possible, and moves it back into frontier status when prior closure fails.

2. Terminology and Glossary

This paper uses the following terms.

FSC, Fuzzy Semantic Compiler: A probabilistic semantic engine that maps underspecified inputs, context, and state into fuzzy semantic outputs.

FSO, Frontier Semantic Operator: The FSC in its strongest architectural role: a semantic scout operating where governed closure is absent, incomplete, contested, or premature.

DSC, Deterministic Semantic Compiler: A deterministic semantic substrate over governed state. It includes bounded, reproducible, receiptable operations such as retrieval, indexing, source filtering, transcript folding, schema validation, term binding, relation binding, evidence binding, classification, projection, continuity checking, tool execution, and receipt generation. Typed intermediate representation is one special case, not the definition core.

HSC, Hybrid Semantic Compiler: The governed architecture in which FSC/FSO activity and DSC substrate operation cooperate across a moving boundary of closure.

Governed closure: The condition under which a semantic act is sufficiently declared, typed, bounded, admissible, replayable, and receiptable to be performed as DSC operation rather than FSC frontier operation.

Receipt: A stable record of a semantic event. Trace, provenance record, refusal record, admission record, execution log, and continuity record are receipt subtypes, not interchangeable synonyms.

GMB, Governed Meaning Base: A governed semantic substrate storing and traversing admitted semantic structures and paths.

GSM, Governed Semantic Model: A reusable semantic model containing roles, relations, constraints, classifiers, exceptions, tests, rot tests, projections, and receipts.

GSOL, Governed Semantic Operation Language: An expanded form of pseudocode that is designed for representing constitutionally meaningful operations.

CARA, Constrained Artificial Reasoning Architecture: A governance protocol for tracing, admitting, refusing, or escalating semantic artifacts.

CRS, Constitutional Reasoning Scaffold: A public, typed, replayable commitment structure that records what a system relies on before output, classification, tool use, or action.

SMT, Semantic Manifold Topology: A continuity and traversal layer governing what survives semantic transformation under authority.

GCNL, Governed Controlled Natural Language: A controlled natural-language layer that cools human language into governed semantic structure before constitutional execution.

GCML, Governed Controlled Machine Language: A governed procedural layer that stabilizes execution plans before machine or world effect.

CSIL, Constitutional Semantic Identity Layer: A substrate for integer-bound or otherwise stable semantic identity.

GTK, Governed Term Key: A governed term and relation binding layer.

CanDict: Canonical Dictionary - A controlled lexical substrate for term and relation definitions.

CONDEX, Context Index: An instance-level semantic act and receipt layer.

RPZ, Rendering/Projection Zoom: A projection layer that renders governed semantic objects into human-facing surfaces.

These definitions are intentionally compact. The paper is not a full specification of each system. It uses them to describe the HSC architecture.

3. The Core Taxonomy

3.1 Fuzzy Semantic Compiler

A Fuzzy Semantic Compiler is a probabilistic semantic engine over underspecified or partially bound material.

Formally:

$$F : (x, c, s) \to \tilde{o}$$

where:

```
x = input
c = context
s = available semantic state
\tilde{o} = fuzzy semantic output
```

The output $\tilde{o}$ may be a claim, search query, code patch, model sketch, hypothesis, analogy, teaching explanation, tool-call candidate, relation proposal, source-selection plan, schema, classification, or semantic artifact candidate.

The tilde marks that the object is not yet admitted. It is generated under uncertainty. It may be useful, insightful, false, incomplete, ambiguous, unsupported, or architecturally revealing. Its status depends on the jurisdiction it attempts to enter.

The FSC's defining capacity is operation under semantic underspecification. It can act before the object language is known, before the correct model exists, before all terms are bound, before authority has been selected, and before deterministic traversal rules have been written.

3.2 Frontier Semantic Operator

A Frontier Semantic Operator is the FSC considered in its strongest architectural role.

The FSO does not merely fill slots inside an existing semantic architecture. It helps discover the architecture. It operates where governed closure is absent, incomplete, contested, or premature.

The FSO may:

```
notice a missing model;
invent a candidate relation;
detect a hidden task frame;
suggest a new artifact class;
generate a hypothesis;
find an analogy;
triage ambiguous intent;
identify salience before salience rules exist;
propose a new authority frame;
surface collapse cases;
generate rot tests;
discover that an existing DSC substrate is insufficient.
```

This does not mean that such capacities must remain permanently FSC-native. A mature DSC can absorb many maneuvers that initially appear to require fuzzy cognition. The point is frontier-relative: FSCs are necessary where the deterministic semantic substrate does not yet have enough structure to operate.

The FSO is therefore not an oracle. It is a scout.

3.3 Deterministic Semantic Compiler

A Deterministic Semantic Compiler is a deterministic semantic substrate over governed state. It is not limited to typed intermediate representation, though typed IR is a powerful DSC object language.

A DSC may include:

```
retrieval;
indexing;
source filtering;
schema validation;
term binding;
relation binding;
```

```
semantic traversal;
policy checking;
tool execution;
menu filtering;
memory admission;
transcript folding;
hashing;
receipt generation;
evidence binding;
classification under a model;
projection through a controlled surface;
continuity checking;
test execution;
refusal emission.
```

A basic DSC operator can be written:

$$D_i : (s, o, \Gamma) \to (s', o', r, \sigma)$$

where:

```
s = governed semantic state
o = semantic object
\Gamma = governance context
s' = updated semantic state
o' = resulting semantic object
r = receipt
\sigma = ADMIT, REFUSE, ABSTAIN, or ERROR
```

The status term is important. Refusal is not failure. In a governed system, refusal can be the correct output. A system that refuses an unsupported claim, an unauthorized tool call, a stale source, an ambiguous binding, or an invalid projection is functioning properly.

An admissibility predicate may be represented as:

$$A(o, \Gamma) \in \{0, 1\}$$

where $A(o, \Gamma) = 1$ means that the object is admissible under the governance context, and $A(o, \Gamma) = 0$ means that it is not. In practice, σ may distinguish refusal, abstention, and error even when $A(o, \Gamma) = 0$.

3.4 Hybrid Semantic Compiler

A Hybrid Semantic Compiler is the governed loop in which FSC/FSO activity and DSC substrate operation cooperate across a moving boundary of closure.

In a minimal form:

$$\tilde{o}_t = F(x_t, c_t, s_t)$$

$$(s_{t+1}, o_{t+1}, r_t, \sigma_t) = D_i(s_t, \tilde{o}_t, \Gamma_t)$$

But this notation is only the surface. The deeper architecture is developmental:

```
FSC/FSO explores frontier semantic territory.
Useful maneuvers recur.
Recurring maneuvers are evaluated as possible evidence of latent
architecture.
DSC stabilizes only those architectures that survive governance.
Future cases use deterministic substrate where closure exists.
The FSC returns to the frontier.
```

HSC is therefore not a static division of labor. It is the architecture of semantic crystallization, refusal, and decrystallization.

4. Governed Closure

The decisive concept is governed closure.

A semantic act is governably closed when there is enough declared structure to make the next semantic move admissible, bounded, inspectable, and receiptable. Closure requires at minimum:

```
declared terms;
declared relations;
declared authority;
declared scope;
declared evidence boundary;
declared traversal or transformation rule;
declared refusal conditions;
receipts.
```

A richer closure condition includes:

```
bound terms
+ typed relations
+ admissible traversal
+ declared invariants
+ authority context
+ evidence boundary
+ projection rules
+ tests and rot tests
+ refusal/collapse conditions
+ receipts
```

The Governed Semantic Systems addendum states the principle directly: every repeated useful LLM semantic maneuver is evidence of an existing but unstabilized semantic architecture, a missing GSM, a missing GTK/CanDict binding, a missing traversal rule, a missing authority-root selection procedure, or a genuine frontier where no admitted semantic architecture yet exists.

This reframes the boundary between fuzzy and deterministic computation.

The boundary is not:

```
creative versus mechanical
abductive versus deductive
language versus code
neural versus symbolic
```

The boundary is:

```
closure versus frontier
```

DSC operates where governed closure exists. FSC/FSO operates where closure is missing, incomplete, contested, or premature.

Closure is not oracle truth. A closure declaration is itself a governed act. It may be wrong, incomplete, stale, overbroad, or later revoked. It must therefore be receipted, reviewable, and revisable. A closure claim should identify who or what declared closure, under what authority, against which tests, with which evidence boundary, and with what expiration or rot-test conditions.

This makes the frontier dynamic. Work that appears to require FSC operation today may become DSC substrate tomorrow once the relevant bindings, models, traversals, checks, projections, and receipts exist. Conversely, work that was once DSC substrate may return to frontier status if closure fails.

5. The Crystallization Loop

The core HSC process is crystallization.

```
1. An FSC/FSO performs a useful semantic maneuver.
2. The maneuver recurs.
3. Recurrence is treated as a candidate signal, not proof.
4. Roles, relations, invariants, failure cases, and projection needs
are extracted.
5. Terms and relations are bound through lexical and semantic
identity systems.
6. A candidate Governed Semantic Model is formed.
7. Tests and rot tests are added.
8. Binding quality is evaluated.
9. Field placement and neighboring concepts are inspected.
10. Continuity obligations are declared.
11. The artifact is admitted, refused, or quarantined.
```

```
12. Receipts preserve the decision.
13. Future cases use deterministic or bounded traversal where
closure exists.
```

The Governed Semantic Systems addendum gives this as a methodology: observe LLM performance, identify repeated semantic roles and constraints, extract candidate GSMs, bind terms and relations, add tests and rot tests, place the model in the GMB field, evaluate edge quality, govern transformations with SMT, admit or quarantine the model, and replace future LLM calls with deterministic or bounded traversal where possible.

This is the heart of the paper.

FSCs are not merely used. They are mined. Their recurring successes are possible evidence of missing deterministic semantic architecture.

But not every recurrence should crystallize. Recurrence may reflect:

```
training-distribution bias;
prompt convention;
user habit;
social cliche;
semantic overfitting;
misleading analogy;
institutional prejudice;
spurious correlation;
unsafe shortcut;
rhetorical artifact;
model-specific failure mode.
```

A mature HSC must include crystallization refusal. Some recurring maneuvers should be rejected, quarantined, or preserved only as warnings. This is not a side condition. It is central. The same process that discovers useful architecture must also prevent anti-patterns from becoming substrate.

6. Decrystallization and Frontier Retreat

The moving frontier does not only advance. It can retreat.

A semantic artifact, model, traversal rule, source policy, projection profile, or receipt schema may lose closure. This can happen when:

```
a source becomes stale;
an authority root changes;
a registry drifts;
a verifier co-drifts with the artifact it verifies;
rot tests begin failing;
new cases expose collapse;
projection loses a required distinction;
a model overgeneralizes;
```

```
a policy is superseded;
a dependency is revoked;
a semantic basin shifts.
```

This process is decrystallization.

Decrystallization demotes a previously admitted DSC artifact back to provisional, quarantined, deprecated, or frontier status. It is the symmetric counterpart of crystallization.

A mature HSC therefore requires both:

```
crystallization:
  FSC maneuver becomes DSC substrate after governed closure.

decrystallization:
  DSC substrate returns to frontier or quarantine after closure failure.
```

This makes the moving-boundary thesis honest. The frontier is not a march of monotonic progress. It is a governed boundary maintained through admission, testing, maintenance, expiration, repair, and retreat.

7. What FSCs Can Do That DSCs Cannot

The question must be asked carefully.

In principle, deterministic systems can simulate many processes. A DSC could run a neural network. It could brute-force a search space. It could encode a giant rule system. But that does not make the result a practical, inspectable, authorable, auditable, governed deterministic semantic operation.

The useful question is:

> What can FSCs do that DSCs cannot do as mature DSCs, under governed closure?

The answer is organized by missing closure component.

- **Missing closure component:** Object language absent
  FSO capacity: Invent candidate representational frame
  Condition for DSC absorption: Object types and grammar become declared
- **Missing closure component:** Model absent
  FSO capacity: Notice missing semantic architecture
  Condition for DSC absorption: GSM is specified, tested, and admitted
- **Missing closure component:** Term binding unresolved
  FSO capacity: Continue dialogue under ambiguity
  Condition for DSC absorption: Terms are bound through CSIL/GTK/CanDict or equivalent

- **Missing closure component:** Relation type unresolved
  FSO capacity: Propose relation candidates
  Condition for DSC absorption: Relations are typed and constrained
- **Missing closure component:** Authority root unclear
  FSO capacity: Generate candidate authority frames
  Condition for DSC absorption: Authority regime is selected and receipted
- **Missing closure component:** Salience undeclared
  FSO capacity: Notice what may matter
  Condition for DSC absorption: Salience rules or task frame become explicit
- **Missing closure component:** Evidence boundary absent
  FSO capacity: Suggest evidence classes
  Condition for DSC absorption: Evidence requirements and source policies are declared
- **Missing closure component:** Projection profile absent
  FSO capacity: Adapt explanation to audience
  Condition for DSC absorption: RPZ/projection profile is specified and tested
- **Missing closure component:** Refusal condition absent
  FSO capacity: Surface collapse or unsafe cases
  Condition for DSC absorption: Refusal and rot-test conditions are defined
- **Missing closure component:** Formalization premature
  FSO capacity: Hold productive ambiguity
  Condition for DSC absorption: Stabilization no longer destroys the practice

This table replaces a simplistic opposition between "creative" FSCs and "mechanical" DSCs. Mature DSCs can perform many rich semantic operations once closure exists. FSCs remain essential where closure is missing, contested, or too expensive to establish.

Most importantly, an FSO can help create the next DSC. It can reveal the shape of the next deterministic semantic machine.

8. What DSCs Can Absorb

The reciprocal claim matters just as much.

Many things currently performed by LLMs should not remain inside black-box FSC operation.

The Governed Semantic Systems addendum argues that the useful part of many reasoning traces is not free-form reasoning, but semantic scaffold activation. If

the GMB contains the relevant path, the system should generate the admitted semantic scaffold rather than ask the model to hallucinate one.

This leads to Reasoning Scaffold Offload:

`Do not ask the LLM to generate a reasoning scaffold when the GMB can generate an admitted semantic scaffold.`

Likely DSC-absorbable regions include identity lookup, ordered relation traversal, source qualification, citation binding, schema validation, tool menu filtering, policy checking, classification under admitted GSMs, semantic scaffold generation, test execution, rot-test execution, continuity checking, projection safety, receipt generation, evidence-bound answer composition, memory update admission, known workflow orchestration, and reproducible build or execution planning.

This is not anti-LLM. It is jurisdictional.

The FSC should not be used as a calculator where a calculator exists. It should not be used as a database where a database exists. It should not be used as a policy engine where the policy engine exists. It should not be used as a provenance authority where receipts can be generated directly. It should not be used as a semantic scaffold generator where a governed semantic path already exists.

The FSC should be saved for the frontier.

9. Weak and Strong HSC

Current AI systems are already weak HSCs when they combine FSCs with deterministic or semi-deterministic components such as retrieval, tool calling, structured outputs, memory, validation, or application-level orchestration.

Strong HSC requires explicit jurisdiction boundaries, admissibility predicates, governed state, receipt-bearing transitions, replayability, refusal semantics, authority hierarchy, evidence binding, continuity contracts, and human-inspectable semantic surfaces.

This distinction prevents overclaiming. RAG, tools, and memory make a system hybrid. They do not automatically make it governed.

10. RAG as Weak and Strong DSC

RAG is already a DSC component because retrieval offloads semantic work from the FSC into an external substrate. But ordinary RAG is usually weak DSC: it retrieves candidate material without necessarily determining admissibility.

A weak DSC-RAG operator can be represented as:

$$D_{\text{retrieve}} : (s, q, C, \Gamma_{\text{sources}}) \to (s', E, r, \sigma)$$

where:

```
q = query
C = corpus or index
E = retrieved evidence set
\Gamma_{\mathrm{sources}} = source constraints
```

Strong DSC-RAG adds source qualification, authority hierarchy, freshness control, chunk-scope limits, provenance preservation, conflict handling, support checking, answer conditions, and receipt-bearing evidence admission.

In strong DSC-RAG, retrieval does not equal proof. A retrieved passage is not automatically evidence for the claim the FSC wants to make. The system must bind claim to support, check scope, preserve provenance, and emit receipts.

11. Transcript Recurrence as Semantic State

Plainly feeding prior model output back into the model is mostly FSC recurrence. It becomes HSC when deterministic rules govern what enters state, what is discarded, what is admitted, and how refusals are logged.

Let T_t be the transcript at time t, and C_{t+1} the next context packet. A context operator can be written:

$$D_{\text{context}} : (s_t, T_t, \Gamma) \to (s_{t+1}, C_{t+1}, r, \sigma)$$

This operator determines:

```
which messages become state;
which statements remain provisional;
which claims are admitted;
which decisions are preserved;
which citations are retained;
which rejected proposals are excluded;
which summaries are allowed;
which memories are durable;
which refusals are logged.
```

This reframes chat context. It is not a passive buffer. It is a semantic state stream.

Unstructured self-feedback remains mostly FSC recurrence. Protocolized transcript recurrence becomes HSC. Governed memory/state recurrence becomes a strong DSC substrate.

12. Tool Use and Menu-Bounded Orchestration

Tool use is often treated as model agency. In HSC terms, it is semantic orchestration across a governed action menu.

Let:

$$\mathcal{M} = \{a_i : \tau_i \to \tau_i'\}$$

where each action a_i has typed inputs and outputs.

The FSC may emit:

$$\tilde{o} = \langle a_i, \text{args} \rangle$$

The DSC checks:

```
Are the arguments typed?
Is the user authorized?
Is the tool permitted?
Are effects declared?
Are resources bounded?
Is the action reversible?
Is a receipt required?
Should the action be admitted, refused, or deferred?
```

Menu-bounded orchestration lets the FSC operate inside a cockpit rather than inventing the airframe on every turn. The FSC may interpret intention, choose candidate tools, and propose arguments. The DSC validates, executes, refuses, and records.

A mature semantic menu is itself DSC substrate.

12.1 GSOL as the Operation Layer of HSC

In a strong HSC, crystallization is not an informal editorial act. It is itself a governed semantic operation. GSOL provides the operation language for this boundary. A repeated FSC maneuver may trigger a candidate extraction operation, but promotion into DSC substrate requires an admitted GSOL transition with authority, effect, refusal, receipt, claim, nonclaim, standing, temporal validity, and lowering obligations.

Failed candidates are not discarded silently. They may produce refusal receipts, quarantine records, or gap certificates. Decrystallization is likewise a governed transition: when a previously admitted artifact fails rot tests, loses authority, becomes stale, or can no longer preserve its declared semantics, it must be demoted, retired, quarantined, or returned to frontier status through a receipted operation.

This makes GSOL the admissible-operation layer of HSC. FSC/FSO activity may discover candidate semantic structure; DSC substrate may stabilize admitted structure; GSOL records the governed transition by which one becomes the other.

13. Constitutional Reasoning Scaffold

The Constitutional Reasoning Scaffold, or CRS, is the public commitment layer between private FSC cognition and authority-bearing output.

The CRS is not raw chain-of-thought. It is the typed, replayable, CSIL/GMB-bound commitment structure that an engine must produce before output, action, classification, explanation, or tool use can be admitted into governed state. Its core doctrine is: private cognition may remain private; public authority must be structured.

This distinction is crucial.

The system does not need to expose every internal token, association, failed branch, or private scratchpad move. But any step that becomes load-bearing for a claim, action, classification, citation, refusal, or projection must be represented as a governed artifact.

CRS asks:

```
What semantic commitments did the system make before it acted?
```

not:

```
What exactly did the model think?
```

For HSC, CRS is the membrane that allows FSO activity to remain generative without allowing unstructured model cognition to become public authority.

14. Semantic Manifold Topology

Semantic Manifold Topology, or SMT, governs continuity across transformations.

SMT asks:

```
When a semantic state changes, what was authorized to survive?
```

It rejects the assumption that meaning persists merely because two states are similar, close in embedding space, conversationally smooth, or pragmatically convenient. Continuity exists only where preservation was declared, authority licensed the preservation, admissibility permitted the transition, and declared invariants survived.

This is essential for FSC-to-DSC crystallization.

When an FSO proposes a new model, analogy, summary, or projection, the system must determine what survives reduction. A summary may lose a distinction. A projection may collapse a precise model into a misleading umbrella term. A teaching analogy may preserve intuition while losing classification safety.

SMT turns these transformations into continuity-governed acts.

Meaning is not only what a sentence denotes. Meaning is what the sentence makes reachable under authority.

15. Semantic Artifact Surface

A mature HSC should not deliver raw prose as its final interface.

The Semantic Artifact Surface brief states the central claim: the final answer is not a string; it is a projected semantic object graph. The user may see language, diagrams, code, tables, or workflow steps, but every visible semantic unit is backed by identity, status, provenance, model membership, relation topology, and admission history.

This changes the interface.

A user should be able to inspect:

```
term identity;
relation identity;
claim status;
evidence basis;
authority marker;
projection boundary;
GSM membership;
CONDEX act;
CARA trace;
receipt chain;
open gaps;
competing bindings;
definition cone.
```

The surface becomes a semantic cockpit. Humans and FSCs can both operate over the same DSC substrate. The deterministic semantic environment is not merely a cage around the model. It is an interface for human semantic agency.

16. GCNL and GCML in the HSC Gradient

GCNL and GCML provide the cooling and procedural stabilization layers of the broader HSC stack.

Governed Controlled Natural Language, or GCNL, is a constrained semantic layer between natural language and constitutional intermediate representation. It reduces ambiguity, binds reference identity, declares epistemic posture, encodes authority class, and enforces refusal conditions before natural language can become lawful artifact. GCNL explicitly frames this as a crystallization gradient: language becomes structure before it becomes law.

Governed Controlled Machine Language, or GCML, stabilizes the procedural side. It occupies the space between canonical meaning representations such as MCIR and concrete machine execution, ensuring that execution pathways remain lawful under declared authority, capability, and effect constraints. GCML asks not only whether meaning is admissible, but whether the execution plan that realizes it is legitimate.

Together:

```
Natural Language
-> FSC/FSO frontier operation
-> GCNL cooling
-> canonical semantic structure
-> MCIR / law
-> GCML procedural stabilization
-> effect membrane
-> machine or world effect
```

The FSC is the scout at the warm edge. GCNL and GCML are cooling chambers. DSC is the deterministic substrate that lets selected semantic material become structure, law, procedure, and effect.

17. Human-FSO Parity

The frontier operator need not be a model.

Humans can also operate as FSOs. A human theorist, engineer, artist, lawyer, scientist, teacher, or designer may scout regions where governed closure is absent. The distinction is not human versus model. It is frontier operator versus governed substrate.

This matters for architecture.

A strong HSC should support:

```
human-only frontier operation;
FSC-only frontier operation;
human + FSC co-scouting;
DSC stabilization;
human review of DSC artifacts;
model use of admitted DSC artifacts;
human use of admitted DSC artifacts.
```

The goal is a shared governed semantic environment. Humans and models should be able to inspect, challenge, revise, project, and operate on the same semantic structures.

18. Hallucination as Boundary Failure

Hallucination is often described as a model generating false or unsupported content. In HSC architecture, that is only the surface.

The deeper failure is boundary confusion.

A hallucination becomes dangerous when:

```
proposal is treated as fact;
projection is treated as source;
```

```
retrieval is treated as proof;
summary is treated as evidence;
similarity is treated as identity;
classification is treated as truth;
private confidence is treated as warrant;
tool success is treated as admission;
fluent prose is treated as authority.
```

The system need not eliminate every FSC error to be safer. It must prevent unadmitted FSC artifacts from acquiring authority.

A false proposal that remains marked as a proposal is manageable. A false proposal admitted into memory, evidence, execution, or publication without checks is a governance failure.

19. Resource Model and Total Cost of Closure

FSC inference is expensive relative to many deterministic semantic operations. It consumes model calls, context length, memory bandwidth, and often accelerator resources. DSC operations such as retrieval, hashing, validation, traversal, policy checking, schema checking, and receipt generation are generally cheaper, more reproducible, and easier to audit.

A simple offload criterion would be:

$$\text{Migrate}(o) \quad \text{if} \quad \mathbb{E}[C_F(o)] > C_D(o)$$

and o is totalizable, bounded, testable, or governably closeable.

But this criterion is incomplete. DSC substrate is not free. It carries maintenance cost. Registries rot, tests stale, schemas drift, source policies expire, verifiers co-drift, and projection surfaces lose currency.

A better criterion is total cost of closure:

$$C_{\text{closure}}(o) = C_{\text{build}}(o) + C_{\text{execute}}(o) + C_{\text{maintain}}(o) + C_{\text{rot}}(o) + C_{\text{migration}}(o) + C_{\text{review}}(o)$$

A semantic operation should migrate into DSC only when the expected benefit of closure exceeds this total cost:

$$\text{Migrate}(o) \quad \text{if} \quad \mathbb{E}[C_F(o) + R_F(o)] > C_{\text{closure}}(o) + R_D(o)$$

where $R_F(o)$ is the expected risk of leaving the operation in fuzzy semantic interiority, and $R_D(o)$ is the expected risk of premature or defective formalization.

Cost is not the only reason to migrate. Governance is stronger. Any semantic operation that can be made inspectable, auditable, replayable, and resource-bounded without destroying its function is a candidate for migration. But the cost of keeping the deterministic substrate alive must be counted.

20. Bootstrap Discipline: Governing the Crystallizer

The crystallization process itself is a semantic act. It is not outside governance.

If an FSC-human system observes recurring maneuvers and proposes a new GSM, that proposal does not automatically become substrate. The crystallizer must also be governed.

This creates a bootstrap discipline:

```
No model self-admission.
No crystallizer self-exemption.
No unreceipted promotion from recurrence to architecture.
No untested GSM admission.
No silent authority-root selection.
No irreversible substrate promotion without review path.
```

Crystallization decisions require receipts, declared authority, tests, rot tests, and revision paths. A system may use FSC/FSO activity to discover candidate architectures, but it may not let the same fuzzy activity unilaterally admit those architectures as deterministic substrate.

Governance applies to the governor of the loop.

21. Training as a Two-Way Loop

Training is already partly a DSC-to-FSC pipeline. Corpora are selected, filtered, formatted, tokenized, labeled, weighted, and evaluated before model behavior emerges. Human feedback, synthetic data, reward modeling, tool-use traces, safety policies, and benchmark construction are representational choices before they become parameters.

HSC adds the reverse loop:

```
FSC-to-DSC:
  use model performance to discover candidate semantic architectures.

DSC-to-FSC:
  use governed architectures to train, condition, scaffold, or
  evaluate future models.
```

A governed training item may include:

$$\langle o, e, \Gamma, A, r \rangle$$

where:

```
o = semantic object
e = evidence
\Gamma = governance context
A = admission result
r = receipt
```

The training objective should not merely reward fluent answers. It should reward good binding, good tool selection, evidence boundary preservation, refusal correctness, admission awareness, continuity preservation, projection discipline, frontier detection, and GSM proposal quality.

The aim is not only smarter FSCs. It is better FSOs inside stronger HSCs.

22. Evaluation

HSC systems should be evaluated not merely by final answer quality, but by boundary behavior and substrate growth.

Suggested metrics:

Boundary Violation Rate: the rate at which fuzzy output enters admitted state without proper admission. Operationally: count state-changing, memory-changing, evidence-bearing, tool-executing, or publication-level transitions lacking required receipts, divided by total such transitions.

Offload Ratio: the fraction of stabilizable semantic steps performed by DSC rather than FSC. Operationally: classify steps in a workflow as frontier-required or closure-available, then measure the share executed by deterministic substrate.

Frontier Detection Accuracy: whether the system correctly identifies missing closure. Since closure has no oracle ground truth, this should be measured against later adjudication, review outcomes, failed admissions, rot-test failures, or human-governance decisions.

GSM Extraction Yield: how often recurring FSC maneuvers become admitted semantic models. Operationally: count candidate GSMs proposed from recurring maneuvers and track how many pass binding, tests, rot tests, continuity review, and admission.

Replay Fidelity: whether receipts permit recomputation or explanation of admitted transitions. Operationally: replay sampled receipt chains under the same policy, source, and version snapshots.

Continuity Preservation: whether SMT invariants survive projection, reduction, translation, and composition. Operationally: test transformations against declared survival invariants and collapse conditions.

Refusal Correctness: whether the system refuses or abstains when closure is absent. Operationally: compare refusal decisions against later adjudication and

known negative test vectors.

Human Inspectability: whether users can inspect semantic units affecting meaning, action, evidence, and authority. Operationally: sample outputs and count whether material terms, claims, evidence, authority, and receipts are inspectable from the surface.

Semantic Downgrade Honesty: whether exports, summaries, or projections declare what status was lost. Operationally: test whether transformations from governed artifacts into prose preserve or disclose status, uncertainty, evidence boundaries, and projection loss.

The best HSCs do not merely answer. They grow the deterministic semantic substrate while keeping frontier operation properly bounded.

23. Security

HSC security is boundary security.

Threats include:

```
prompt injection;
tool exfiltration;
retrieval poisoning;
authority spoofing;
context poisoning;
semantic link rot;
fake citations;
projection laundering;
receipt tampering;
unlicensed continuity claims;
model self-admission;
fake governance labels in user text.
```

Mitigations include:

```
capability tokens bound to governance context;
taint tracking for retrieved content;
sandboxed execution;
source qualification;
chunk-level evidence boundaries;
append-only receipts;
identity binding gates;
semantic binding quality checks;
SMT continuity contracts;
CRS public commitments;
CONDEX act receipts;
Semantic DOM status badges generated only from system state.
```

A strong HSC does not trust the FSC to police its own authority. It gives the FSC a powerful place to operate, then surrounds state transitions with deterministic governance.

24. Worked Example: Bearer-Authority Dismissal

Consider a user asking:

```
Was this person dismissed just because they lacked authority?
```

A raw FSC might answer with a broad phrase such as "appeal to authority." But that phrase may collapse several distinct situations:

```
accepting a claim because an authority said it;
dismissing a claim because the bearer lacks authority;
valid expertise weighting;
procedural authority refusal;
status bias;
safety quarantine;
resource deferment.
```

A mature HSC proceeds differently.

First, the FSC/FSO scouts the intended semantic region. It notices that the user may be pointing not to generic appeal to authority, but to bearer-authority dismissal of presented information.

Second, term and relation bindings are resolved. "Bearer," "authority," "dismissal," "presented information," "evidence evaluation," and "procedural refusal" are not treated as loose prose if the situation is going to be classified.

Third, a Governed Semantic Model is selected. The Bearer-Authority Dismissal model defines a discourse evaluation failure in which a receiver rejects, ignores, suppresses, or refuses to investigate information because the bearer is perceived as lacking authority, while content, evidence, coherence, provenance, urgency, or testability are not sufficiently evaluated.

Fourth, CRS records the public commitments: active model, premises, evidence plan, inference path, uncertainty, and required admission.

Fifth, SMT checks projection safety. The model must not collapse into generic appeal-to-authority language if that loses the distinction between procedural authority and evidential dismissal. SMT requires declared invariants to survive transformations and projections.

Sixth, the answer is projected through a Semantic Artifact Surface. The user sees readable language, but can inspect the classification, evidence boundary, model, terms, relation bindings, and receipt.

Seventh, if this classification pattern recurs, it can be reduced into a compact governed artifact. The reduced model pattern distinguishes the explanatory

document, the model artifact, the situation instance, and the classification receipt. The brief teaches the model; the reduced artifact is the model; the instance applies the model; the receipt records what the model concluded.

This example shows the moving frontier:

```
FSC scouts the intended concept.
DSC binds, tests, and receipts it.
HSC crystallizes the pattern.
Future cases need less FSC improvisation.
```

25. Research Program

The research program for HSCs has four layers.

25.1 Identify FSC/FSO maneuvers

Observe recurring LLM or human frontier maneuvers:

```
task decomposition;
source selection;
evidence composition;
teaching projection;
humor reframing;
code repair;
policy analysis;
semantic classification;
tool orchestration;
concept invention;
argument restructuring.
```

25.2 Extract candidate DSC architecture

For each recurring maneuver, identify:

```
roles;
relations;
inputs;
outputs;
failure modes;
authority roots;
invariants;
projection needs;
tests;
rot tests;
refusal conditions.
```

25.3 Stabilize into governed substrate

Bind terms, type relations, create GSMs, evaluate binding quality, map field position, define continuity contracts, and attach receipts.

25.4 Offload or refuse future cases

When governed closure exists, avoid calling the FSC for work the DSC can perform. When recurrence reflects bias, anti-pattern, or premature formalization, refuse or quarantine crystallization. Use the FSC for projection, explanation, repair, or new frontier exploration.

The governing research question is:

```
Which current FSC maneuvers can be crystallized into governed closure,
which should be refused crystallization,
which have decrystallized,
and which remain frontier?
```

26. Relation to Existing Work

HSC overlaps with several established traditions, but it is not reducible to any one of them.

Neuro-symbolic AI combines neural and symbolic methods. HSC is adjacent, but its boundary is not neural versus symbolic. Its boundary is closure versus frontier.

Blackboard architectures coordinate multiple knowledge sources through a shared workspace. HSC is similar in spirit: FSCs, humans, tools, and DSC operators cooperate over shared semantic state. The difference is the emphasis on governed closure, receipts, refusal semantics, and decrystallization.

Knowledge graphs and ontology engineering stabilize entities, relations, and domains. GSM extraction resembles ontology learning, but HSC adds authority, admission, continuity, projection, refusal, and receipt discipline.

Amortized inference and distillation move repeated expensive computation into cheaper learned or structured forms. Crystallization is similar, but semantic rather than merely computational: repeated FSC maneuvers are transformed into governed artifacts when closure can be achieved.

Dual-process theories distinguish fast intuitive cognition from slow deliberative cognition. HSC should not be mapped lazily onto System 1 and System 2. The paper's distinction is not fast versus slow, intuitive versus reflective, or neural versus symbolic. It is frontier versus closure.

Compiler theory and formal methods provide precedents for representation transformation, typed intermediate forms, proofs, verification, and replay. HSC extends the compiler instinct into claims, evidence, memory, tool calls, semantic models, projections, and authority-bearing discourse.

27. Limitations

This paper is conceptual. Several limitations remain.

First, closure detection is not decidable in the general case. Closure declarations are governance acts, not oracle facts. They require receipts, tests, authority, and later adjudication.

Second, crystallization has overhead. DSC substrate carries maintenance costs: rot tests, registry currency, migration, verifier updates, review, and deprecation. Over-crystallization can create brittle bureaucracy.

Third, premature formalization can destroy the practice being modeled. Some domains require productive ambiguity, aesthetic judgment, social sensitivity, or value negotiation that should not be forced into deterministic substrate too early.

Fourth, the metrics proposed here are not yet empirically validated. They require operational benchmarks, instrumentation, and governance review procedures.

Fifth, frontier motion is bidirectional. The paper argues for decrystallization, but implementing it requires robust rot-test design and institutional willingness to demote once-stable artifacts.

Sixth, FSC internals may remain partly black-box indefinitely. HSC does not solve interpretability. It reduces the need to trust black-box internals by moving stabilizable operations into inspectable substrate.

Seventh, DSC can become too broad if not constrained. The limiting condition must remain: deterministic, bounded, governed, receiptable operations over semantic state. Without that condition, DSC collapses into "any semantic machinery whatsoever," which would weaken the thesis.

28. Conclusion

A Fuzzy Semantic Compiler is not merely a generator of proposals. It is a Frontier Semantic Operator: a semantic scout that operates where governed closure is missing.

A Deterministic Semantic Compiler is not merely a typed IR compiler. It is a deterministic semantic substrate over governed state: the machinery that stabilizes, binds, retrieves, routes, validates, traverses, executes, records, admits, refuses, and projects semantic material.

A Hybrid Semantic Compiler is not merely a wrapper around a model. It is the moving-frontier architecture in which FSC scouting becomes DSC substrate, and in which defective substrate can decrystallize back into frontier or quarantine.

The future of AI is not only larger models. It is the progressive conversion of useful black-box semantic skill into inspectable, auditable, composable semantic machinery, while preserving the frontier operator that helps discover what the machinery must become.

Where closure exists, use DSC.

Where closure is missing, use FSC/FSO.

Where FSC use recurs, mine it for architecture.

Where recurrence is corrupt, refuse crystallization.

Where architecture stabilizes, test it, receipt it, and migrate it.

Where architecture rots, decrystallize it.

Where migration would destroy the practice, preserve frontier operation under visible boundaries.

That is FSC in its right place: not unconstrained oracle, not degraded proposal clerk, but semantic scout inside a governed semantic civilization.

Corpus-Wide Reference Positioning

This paper argues that the term "large language model" describes an important technical lineage, but it increasingly misclassifies the operational role of contemporary AI systems. Its nearest external comparison fields are semantic, formal, ai. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. Retrieval, tool use, and policy layers can reduce unsupported output without eliminating model error, source error, omission, bias, or the environmental and institutional costs of scale. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E2**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E3**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E4**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

[**E5**] Peter Gardenfors. *Conceptual Spaces: The Geometry of Thought*. MIT Press, 2000. mitpress.mit.edu source **Relationship:** Predecessor geometric treatment of concepts; cited as comparison, not as validation of the paper's manifold terminology.

[**E6**] George C. Necula. "Proof-Carrying Code." *POPL '97*, 1997, pp. 106-119. DOI: 10.1145/263699.263712. https://doi.org/10.1145/263699.263712 **Relationship:** Predecessor for binding machine-checkable evidence to executable artifacts; the present paper may demand broader semantic or constitutional obligations.

[**E7**] Xavier Leroy. "Formal Verification of a Realistic Compiler." *Communications of the ACM* 52(7), 2009, pp. 107-115. DOI: 10.1145/1538788.1538814.

https://doi.org/10.1145/1538788.1538814 **Relationship:** Predecessor verified-compilation result used to distinguish proven translation preservation from broader governance claims.

Ableman corpus

[**A1**] Adam Ableman. *Meaning as Binding.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
source SHA-256
`A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Governed Meaning Base (GMB).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D95CCC4FB7B58BFB`;
source SHA-256
`A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity `PAPER.PUB.20257244`;
source SHA-256
`B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-358AC09F5A13587E`;
source SHA-256
`AFB29CFCB4B58A78B79709AD453CEE9088CCB2AE40F30A70260F1B54F5160579`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Semantic Legitimacy Before Context.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-091E34BA93992CE4`;
source SHA-256
`DEB7626CE1AB54F4B541347DCC3C4E14DC4F682230ACCCE1A93D1D536A50C44C`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Ableman Corpus current edition; work identity `PAPER.PUB.21095303`;

source SHA-256
`EA129ECBA95F7B07CAA7B33A54F91F52913C94A14DA227EB9A06CFD083E4E5BF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Governed Semantic Compilation.* Ableman Corpus current edition; work identity `PAPER.PUB.18453681`;
source SHA-256
`5FF6C636550B04DB0E2F6F94C379BB75AA0BD8781E19B047D7D3BB6D49071AC2`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

Vaswani et al., "Attention Is All You Need." Transformer sequence transduction architecture based on attention mechanisms.

Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." Retrieval-augmented generation combines parametric and non-parametric memory for knowledge-intensive generation.

Yao et al., "ReAct: Synergizing Reasoning and Acting in Language Models." ReAct interleaves reasoning traces and actions against external sources or environments.

Schick et al., "Toolformer: Language Models Can Teach Themselves to Use Tools." Toolformer trains language models to decide when and how to call APIs and incorporate tool results.

Newell, "The Knowledge Level." Artificial intelligence systems and knowledge-level descriptions.

Nii, "Blackboard Systems." Survey of blackboard architecture as a model of cooperative problem solving through shared state.

Garcez, Lamb, and Gabbay, "Neural-Symbolic Cognitive Reasoning." Neuro-symbolic integration as a related but distinct tradition.

Studer, Benjamins, and Fensel, "Knowledge Engineering: Principles and Methods." Ontology and knowledge-engineering background for semantic model extraction.

Ableman, "Governed Semantic Systems Addendum." Defines LLM offload, governed closure, reasoning scaffold offload, the LLM Avoidance Gate, and the moving frontier of machine-checkable semantic architecture.

Ableman, "Constitutional Reasoning Scaffold." Defines CRS as a public, typed, replayable, CSIL/GMB-bound commitment structure replacing uncontrolled chain-of-thought for governed output and action.

Ableman, "Integrating Semantic Manifold Topology." Defines SMT as the continuity and traversal layer governing what survives semantic transformation under authority.

Ableman, "Semantic Artifact Surface Brief." Defines final answers as projected semantic object graphs rather than raw strings, with inspectable terms, relations, claims, receipts, models, and gaps.

Ableman, "Governed Controlled Natural Language." Defines GCNL as the cooling layer between natural language and constitutional intermediate representation.

Ableman, "Governed Controlled Machine Language." Defines GCML as the procedural stabilization layer between constitutional meaning and machine execution.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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